```
import numpy as np
import pandas as pd
import matplotlib as mpl
import matplotlib.pyplot as plt
import seaborn as sns
import yfinance as yf
plt.rcParams['figure.dpi'] = 150
plt.rcParams['axes.grid'] = True
plt.rcParams['axes.spines.top'] = False
plt.rcParams['axes.spines.right'] = False
%matplotlib inline
%pandas_setup -num_digits=4 -max_rows=20 -max_cols=20
%load_ext autoreload
%autoreload 2
%load_ext rpy2.ipython
%R library(ggplot2)
%R library(dplyr)
%R library(tidyverse)
%R library(lubridate)
%R library(zoo)
%R library(tsibble)
%R library(tsibbledata)
%R library(feasts)
%R library(fable)
%R library(fabletools)
%R library(ggfortify)
%R theme_set(theme_bw())
%R options(tibble.width = Inf)
%R options(dplyr.print_max = Inf)
%R options(dplyr.summarise.inform = FALSE)
%R options(lubridate.verbose = TRUE)
%R options(warn = -1)
%R options(repr.plot.width=8, repr.plot.height=4)
```

1 - Realized Volatility

We provide Pyhton Pandas based `pandas_datareader` to no longer supported for better `QUANT` implementations.

```
pd.pandas_datareader.yahoo.daily.YahooDailyReader(
    symbols='^GSPC',
    start=pd.Timestamp('2000-01-01'),
    end=None,
    interval='d',
    chunksize=5000,
    retry_count=3,
    pause=0.1,
    session=None,
    adjust_prices=False,
    ).read().sort_index().
    ).read().sort_index().
    ).read().sort_index().
```